

Prescribed Isolated Intervals in Higher Fold Sumsets

Zijie Yu
Ningbo University
Ningbo, China
256001428@nbu.edu.cn

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Abstract

We give an explicit construction of a finite set of integers whose h -fold sumset contains any prescribed finite target set and whose remaining elements are isolated. As a consequence, every sufficiently separated family of intervals can be prescribed as exactly the nontrivial intervals in an h -fold sumset. This answers the public multi-interval questions numbered Problems 14 and 15 in version 3 of Nathanson's *Problems in additive number theory, VII*. In fact the required gap threshold is the minimal value $N_q = n + 2$ allowed by the formulation, and the construction has $|A| = 2q(n + 1)$.

The proof uses a separated-label principle that reduces the construction to excluding short integer relations among labels. This framework also gives quantitative diameter refinements: finite moment labels give polynomial diameter for fixed h , Bose–Chowla labels improve the fixed- h exponent, and shifted-Sidon labels give a quadratic double-sumset bound with a matching Sidon-method lower bound.

Finally, we study cardinality separately from complete isolation. A two-colour bounded Frobenius packet gives a single-interval double-sumset construction with $O(\sqrt{n})$ elements, which is optimal in order. Under an additional pair-sum separation hypothesis on the starts, the same packet gives an $O(q\sqrt{n})$ starts-only multi-interval construction. A second sparse multi-interval construction uses Chinese-remainder incompatible colours to remove the pair-sum separation hypothesis, at the cost of a q -dependent large- n threshold and larger spacing between the prescribed intervals. The main separation, interval-family, cardinality, translation, diameter, and shifted-Sidon arguments have been checked in Lean 4 with Mathlib.

1 Introduction

Nathanson's problem list on additive h -bases for intervals asks how much control is possible over the intervals contained in finite sumsets. In version 3, posted on June 8, 2026, the single-interval double-sumset case is constructed directly, and the public multi-interval questions, Problems 14 and 15, ask whether one can prescribe a separated set

$$C = \{c_1 < \cdots < c_q\}$$

of starting points so that the only intervals of length n start at the elements of C , with a stronger version forbidding all additional proper intervals. The stated problems allow arbitrary fixed $h \geq 2$.

The point of this note is that the multi-interval problem is more naturally a finite-target problem. Let

$$S = \bigcup_{i=1}^q [c_i, c_i + n].$$

It is enough to construct A so that

$$hA = S \dot{\cup} E,$$

where every point of E is separated from S and from every other point of E by at least two. Then the maximal nontrivial intervals in hA are exactly the maximal nontrivial intervals already present in S . The construction below proves this for every $h \geq 2$ and, more generally, for an arbitrary finite target set $S \subseteq \mathbb{Z}$.

The mechanism is a tagged cancellation construction. To each target point t_r we attach a large label a_r and put two generators into A ,

$$t_r + (h-1)a_r \quad \text{and} \quad -a_r.$$

The intended representation of t_r is the matched sum consisting of one positive generator and $h-1$ copies of the corresponding negative generator. All other h -term sums carry a nonzero label. If the label family has no short relations, then these exceptional sums are far apart.

The first version uses powers and has exponential diameter. The later sections replace the powers by finite moment labels and then by a Bose–Chowla modular B_t construction, giving polynomial diameter for fixed h with a smaller exponent. When $h = 2$, shifted-Sidon labels give a direct quadratic diameter bound. The Sidon section also includes a lower bound showing that this quadratic order is forced by the Sidon-label strategy. These quantitative refinements are not needed merely to answer the existence questions, but they separate the flexible cancellation mechanism from the arithmetic problem of choosing efficient labels.

There is another, complementary optimization in the double-sumset case. If one asks only for the starts of intervals of length n , rather than complete isolation of every other sum, then an interval need not be generated point by point. A two-colour packet built from arithmetic progressions with steps a and $a+1$ produces one long interval in the cross-sum while the two same-colour sumsets have no adjacent elements. This gives an order-optimal sparse single-interval construction with $O(\sqrt{n})$ elements, and an $O(q\sqrt{n})$ multi-interval version whenever the pair sums $c_i + c_j$ are also separated. A further CRT variant gives the same $O(q\sqrt{n})$ cardinality order for arbitrary prescribed starts, provided q is fixed, n is sufficiently large in terms of q , and the prescribed intervals are separated on the scale of $3n$.

2 A short separation lemma

For an integer vector $v = (v_0, \dots, v_{s-1})$, write $\|v\|_1 = \sum_r |v_r|$.

Lemma 1 (Positional separation). *Let $B, M \geq 1$ and $a_r = MB^r$ for $0 \leq r < s$. If $\lambda, \mu \in \mathbb{Z}^s$, $\lambda \neq \mu$, and $\|\lambda\|_1 + \|\mu\|_1 \leq B$, then*

$$\left| \sum_{r=0}^{s-1} (\lambda_r - \mu_r) a_r \right| \geq M.$$

Proof. Put $z = \lambda - \mu$ and let R be the largest index with $z_R \neq 0$. We have $\|z\|_1 \leq B$. If $R = 0$, the result is immediate. If $R \geq 1$, then

$$\begin{aligned} \left| \sum_{r=0}^{s-1} z_r a_r \right| &\geq M \left(|z_R| B^R - \sum_{r < R} |z_r| B^r \right) \\ &\geq M (B^R - (B-1)B^{R-1}) \geq M. \end{aligned}$$

Here the sum of the absolute values of the lower coefficients is at most $B - |z_R| \leq B - 1$. □

Proposition 1 (Separated-label principle). *Let $h \geq 2$, $d \geq 1$, and let $S = \{t_0, \dots, t_{m-1}\} \subseteq \mathbb{Z}$ be nonempty. Put*

$$H = h(h-1), \quad T = \max_{t \in S} |t|.$$

Suppose that $M \geq 2hT + d$ and that integers $a_0, \dots, a_{m-1}$ satisfy

$$\left| \sum_{r=0}^{m-1} (\lambda_r - \mu_r) a_r \right| \geq M \quad (1)$$

whenever $\lambda, \mu \in \mathbb{Z}^m$ are distinct and

$$\|\lambda\|_1 + \|\mu\|_1 \leq 2H.$$

Then

$$A = \{t_r + (h-1)a_r : 0 \leq r < m\} \cup \{-a_r : 0 \leq r < m\}$$

has $|A| = 2m$ and

$$hA = S \dot{\cup} E,$$

where distinct elements of E are at distance at least d from one another and every element of E is at distance at least d from S .

Proof. Attach to the generator $t_r + (h-1)a_r$ the label vector $(h-1)e_r$ and offset t_r . Attach to the generator $-a_r$ the label vector $-e_r$ and offset 0. The integer value of a tagged generator is its offset plus the dot product of its label vector with $(a_0, \dots, a_{m-1})$.

We first check that the displayed union defining A is disjoint and has the claimed cardinality. Distinct tagged generators have distinct label vectors. The sum of the ℓ^1 norms of two generator labels is at most $2(h-1) \leq 2H$, and their offsets differ by at most $2T$. Since $M \geq 2hT + d > 2T$, the separation hypothesis (1) implies that two distinct tagged generators cannot have the same integer value. Hence $|A| = 2m$.

Now consider an arbitrary h -term sum from A . Let p_r be the number of times the positive generator $t_r + (h-1)a_r$ is used, and let u_r be the number of times the negative generator $-a_r$ is used. Thus

$$\sum_r (p_r + u_r) = h.$$

The label coefficient of this representation is

$$v_r = (h-1)p_r - u_r.$$

Its norm is bounded by

$$\sum_r |v_r| \leq (h-1) \sum_r p_r + \sum_r u_r \leq h(h-1) = H,$$

and the offset $\sum_r p_r t_r$ has absolute value at most hT .

The target points are exactly the sums with zero label. Indeed, if $v = 0$, then $(h-1)p_r = u_r$ for every r . Therefore

$$\sum_r (p_r + u_r) = h \sum_r p_r = h,$$

so $\sum_r p_r = 1$. There is exactly one index r with $p_r = 1$, and then $u_r = h-1$ while all other counts vanish. The corresponding sum is

$$(t_r + (h-1)a_r) + (h-1)(-a_r) = t_r.$$

Conversely, this matched representation gives every t_r . Thus the zero-label part of hA is precisely S .

It remains to see that exceptional sums cannot cluster. A nonzero label vector determines its multiset of terms. Suppose two h -term multisets with counts (p_r, u_r) and (p'_r, u'_r) have the same label vector. Then

$$u_r - u'_r = (h-1)(p_r - p'_r).$$

Summing this identity over r and using the fact that both multisets have h terms gives $\sum_r p_r = \sum_r p'_r$. If the positive count vectors were not equal, choose an index with $p_r > p'_r$. Then $p_r \geq 1$ and the displayed identity gives $u_r \geq h - 1$. These already use all h terms, so the first multiset is a matched zero-label representation, contradicting the assumption that the common label vector is nonzero. Hence $p = p'$ and then also $u = u'$.

Consequently, two distinct exceptional sums have distinct nonzero label vectors. Also, every exceptional sum has a label vector distinct from the zero label of a target point. For two exceptional sums, or for an exceptional sum and a target point, the two label vectors have combined norm at most $2H$. Equation (1) gives label separation at least M , while the two offsets together can change the distance by at most $2hT$. The remaining distance is at least

$$M - 2hT \geq d.$$

This proves the required decomposition and isolation. $\square$

3 The construction

Theorem 1 (Finite target realization). *Let $h \geq 2$, $d \geq 1$, and let $S \subseteq \mathbb{Z}$ be a nonempty finite set with $m = |S|$. There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2m$ such that*

$$hA = S \dot{\cup} E,$$

where distinct elements of E are at distance at least d from one another and every element of E is at distance at least d from S .

Proof. Enumerate $S = \{t_0, \dots, t_{m-1}\}$ and put

$$T = \max_{t \in S} |t|, \quad M = 2hT + d, \quad B = 2h(h-1), \quad a_r = MB^r.$$

Lemma 1 verifies (1) for these labels: if the two coefficient vectors have combined ℓ^1 norm at most $B = 2h(h-1)$, then their positional values differ by at least M . Proposition 1 therefore gives the required set

$$A = \{t_r + (h-1)a_r : 0 \leq r < m\} \cup \{-a_r : 0 \leq r < m\}.$$

$\square$

Theorem 2 (Prescribed isolated intervals). *Let $h \geq 2$, $n \geq 1$, $d \geq 2$, $q \geq 1$, and*

$$C = \{c_1 < \dots < c_q\} \subseteq \mathbb{Z}$$

with $c_{i+1} - c_i \geq n + 2$ for all i . There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2q(n+1)$ such that

$$hA = S \dot{\cup} E, \quad S = \bigcup_{i=1}^q [c_i, c_i + n],$$

where

$$|x - y| \geq d$$

whenever $x, y \in E$ are distinct or $x \in E$ and $y \in S$. Consequently, the maximal intervals in hA of length at least one are exactly the intervals $[c_i, c_i + n]$.

Proof. Because $c_{i+1} - c_i \geq n + 2$, the intervals $[c_i, c_i + n]$ are disjoint and separated from one another by at least one missing integer. Hence $|S| = q(n+1)$. Apply Theorem 1 to this finite target set S . It gives $|A| = 2q(n+1)$, the decomposition $hA = S \dot{\cup} E$, and the required distance bounds. Since $d \geq 2$ and the intervals in S are separated by at least one missing integer, no exceptional point can extend a target interval or form a new nontrivial interval. $\square$

Corollary 1 (Single interval specialization). *For all $h \geq 2$, $c \in \mathbb{Z}$, $n \geq 1$, and $d \geq 2$, there is a set $A \subseteq \mathbb{Z}$ with $|A| = 2n + 2$ such that $[c, c + n] \subseteq hA$, every element of $hA \setminus [c, c + n]$ is at distance at least d from every other such element and from $[c, c + n]$, and the longest interval in hA has length n .*

Proof. Apply Theorem 2 with $C = \{c\}$. □

Corollary 2 (Nathanson's public multi-interval problems). *In the notation of Problems 14 and 15 in version 3 of Nathanson's Problems in additive number theory, VII, both problems are true for every $h \geq 2$. More precisely, one can take the threshold*

$$N_q = n + 2$$

and

$$k_C = 2q(n + 1).$$

Proof. The hypotheses in the revised problems include $2 \leq n \leq N_q - 2$, so $N_q = n + 2$ is the smallest possible threshold compatible with that formulation. If $c_{i+1} - c_i \geq n + 2$, then Theorem 2 with $d = 2$ gives

$$hA = S \dot{\cup} E, \quad S = \bigcup_{i=1}^q [c_i, c_i + n],$$

where every point of E is isolated from every other point of hA .

The only intervals of length n in hA are the displayed intervals $[c_i, c_i + n]$, proving Problem 14. Since every element outside S is isolated and at distance at least two from S , every interval of length one in hA lies inside one of the intervals $[c_i, c_i + n]$, proving the stronger Problem 15. □

Remark 1 (Diameter bound). *For*

$$S = \bigcup_{i=1}^q [c_i, c_i + n], \quad T = \max_{s \in S} |s|,$$

the construction in Theorem 2 satisfies

$$\text{diam}(A) \leq h(2hT + d)(2h(h - 1))^{q(n+1)-1} + T.$$

The bound is intentionally simple rather than optimized.

4 A sparse construction for one interval

The construction above proves the stronger no-extra-interval assertion with $2q(n + 1)$ elements. For a single interval in a double sumset, one can do much better in the interval length if shorter proper intervals elsewhere are allowed.

Lemma 2 (A two-colour Frobenius packet). *Let $a \geq 2$, put $b = a + 1$, and let $R \geq b - 1$, $S \geq a - 1$. Define*

$$X = \{0, a, 2a, \dots, Ra\}, \quad Y = \{0, b, 2b, \dots, Sb\},$$

and put

$$g = (a - 1)(b - 1) = a(a - 1), \quad L = aR + bS - 2g.$$

Then $X + Y$ contains the interval

$$[g, g + L].$$

If $L \geq g$, this is the only interval of length L in $X + Y$. Moreover $X + X$ and $Y + Y$ contain no nontrivial intervals.

Proof. The last assertion is immediate: $X + X$ is contained in one residue class modulo a , and $Y + Y$ is contained in one residue class modulo b , with $a, b \geq 2$.

We use the standard bounded form of the two-generator Frobenius argument: if $\gcd(a, b) = 1$, $R \geq b - 1$, and $S \geq a - 1$, then

$$\{ia + jb : 0 \leq i \leq R, 0 \leq j \leq S\}$$

contains every integer in the interval

$$[(a - 1)(b - 1), aR + bS - (a - 1)(b - 1)].$$

For completeness, let N be in this interval. Choose $j_0 \in \{0, \dots, a - 1\}$ with $N - j_0b \equiv 0 \pmod{a}$. Since $N \geq (a - 1)(b - 1)$, the integer

$$i_0 = \frac{N - j_0b}{a}$$

is nonnegative. If $i_0 \leq R$, then $N = i_0a + j_0b$ is already in the box. Otherwise choose the least integer $t \geq 1$ such that $i_0 - tb \leq R$, and put

$$i = i_0 - tb, \quad j = j_0 + ta.$$

Then $i \leq R$, and the minimality of t gives $i > R - b$. Since $R \geq b - 1$, this implies $0 \leq i \leq R$. If $j > S$, then $i \geq R - b + 1$, and therefore

$$N = ia + jb \geq a(R - b + 1) + b(S + 1) = aR + bS - (a - 1)(b - 1) + 1,$$

contradicting the upper bound. Thus $j \leq S$, and $N = ia + jb$ lies in the bounded sumset. Applying this with $b = a + 1$ gives the interval $[g, g + L]$.

The integer $g - 1$ is the Frobenius number for a and b , so it is not in $X + Y$; by symmetry, $aR + bS - g + 1$ is not in $X + Y$ either. All points of $X + Y$ outside $[g, g + L]$ lie in the two side ranges

$$[0, g - 1], \quad [g + L + 1, g + L + g].$$

Each side range has diameter $g - 1 < L$ when $L \geq g$. Therefore no side range contains an interval of length L , and no interval of length L can extend the central interval across either missing endpoint. $\square$

Theorem 3 (Sparse single-interval realization). *Let $n \geq 16$, let $c \in \mathbb{Z}$, and put*

$$a = \lfloor \sqrt{n} \rfloor - 1, \quad g = a(a - 1).$$

Then there is a finite set $A \subseteq \mathbb{Z}$ such that $[c, c + n]$ is the unique interval of length n in $2A$, and

$$|A| \leq 5\sqrt{n} + 1.$$

Proof. Put

$$n_0 = n - (2a - 1).$$

Since $a = \lfloor \sqrt{n} \rfloor - 1$ and $n \geq 16$, we have $a \geq 3$ and $n_0 \geq a^2 - 1$. Choose $v \in \{0, \dots, a - 1\}$ such that

$$v \equiv n_0 \pmod{a},$$

and define

$$u = \frac{n_0 - (a + 1)v}{a}.$$

The inequality $n_0 \geq a^2 - 1$ gives $u \geq 0$. Set

$$R = a + u, \quad S = a - 1 + v, \quad b = a + 1.$$

Then $R \geq b - 1$, $S \geq a - 1$, and

$$aR + bS - 2a(a - 1) = 2a - 1 + au + (a + 1)v = n.$$

Also $g < a^2 \leq n$.

Let X and Y be the two arithmetic progressions from Lemma 2. Choose an integer P so large that the three families of sums displayed below are pairwise separated by at least one missing integer; for example any sufficiently large positive P works. Put

$$A = (P + X) \cup (c - g - P + Y).$$

The cross-sums are

$$(P + X) + (c - g - P + Y) = c - g + X + Y.$$

By Lemma 2, this cross-sum packet has a unique interval of length n , namely $[c, c + n]$. The same-colour sums

$$(P + X) + (P + X) \quad \text{and} \quad (c - g - P + Y) + (c - g - P + Y)$$

contain no nontrivial intervals, since they are translates of $X + X$ and $Y + Y$. The choice of P keeps these same-colour sums away from the cross-sum packet. Hence $[c, c + n]$ is the unique interval of length n in $2A$.

Finally,

$$|A| \leq (R + 1) + (S + 1).$$

Because $v < a$ and $a \leq \sqrt{n}$, we have $S + 1 = a + v \leq 2\sqrt{n}$. Also $a \geq \sqrt{n}/2$ for $n \geq 16$, and hence

$$R + 1 = a + u + 1 \leq a + \frac{n_0}{a} + 1 \leq 3\sqrt{n} + 1.$$

Combining these two estimates gives the claimed bound. $\square$

Proposition 2 (Optimal order for one interval). *Let $A \subseteq \mathbb{Z}$ be finite, put $k = |A|$, and suppose that $2A$ contains an interval of length n . Then*

$$k \geq \frac{\sqrt{8n + 9} - 1}{2}.$$

Consequently, the bound in Theorem 3 has the best possible order of growth in n .

Proof. The sum $a + a'$ is determined by the unordered pair $\{a, a'\}$, with repetition allowed. Hence

$$|2A| \leq \binom{k + 1}{2}.$$

If $2A$ contains an interval of length n , then it contains at least $n + 1$ distinct integers. Therefore

$$n + 1 \leq \binom{k + 1}{2} = \frac{k(k + 1)}{2},$$

which gives the displayed lower bound after solving the quadratic inequality for k . $\square$

Theorem 4 (Sparse pair-separated multi-interval realization). *Let $n \geq 16$, let $q \geq 1$, and define*

$$a = \lfloor \sqrt{n} \rfloor - 1, \quad b = a + 1, \quad g = a(a - 1).$$

Put

$$n_0 = n - (2a - 1),$$

choose $v \in \{0, \dots, a-1\}$ with $v \equiv n_0 \pmod{a}$, and set

$$u = \frac{n_0 - (a+1)v}{a}, \quad R = a + u, \quad S = a - 1 + v.$$

Let

$$C = \{c_1 < \dots < c_q\} \subseteq \mathbb{Z}$$

and suppose that distinct elements of

$$C + C = \{c_i + c_j : 1 \leq i \leq j \leq q\}$$

are separated by at least

$$\Delta = \max(n + 2g + 2, 2Sb + 2).$$

Then there is a finite set $A \subseteq \mathbb{Z}$ such that the intervals $[x, x+n] \subseteq 2A$ are exactly the intervals $[c_i, c_i+n]$, and

$$|A| \leq (2q+3)\sqrt{n} + 1.$$

Proof. As in the proof of Theorem 3, the chosen parameters satisfy

$$R \geq b - 1, \quad S \geq a - 1, \quad aR + bS - 2g = n, \quad n \geq g.$$

Let

$$X = \{0, a, 2a, \dots, Ra\}, \quad Y = \{0, b, 2b, \dots, Sb\}.$$

Choose P so large that the families

$$2P + X + X, \quad c_i - g + X + Y, \quad c_i + c_j - 2g - 2P + Y + Y$$

are separated from families of the other two displayed types by at least one missing integer. This is possible because P moves the first type to the right and the third type to the left, while the cross-sum packets stay fixed. Put

$$A = (P + X) \cup \bigcup_{i=1}^q (c_i - g - P + Y).$$

The cross-sums are

$$(P + X) + (c_i - g - P + Y) = c_i - g + X + Y.$$

By Lemma 2, the i th cross-sum packet contains the unique interval of length n

$$[c_i, c_i + n].$$

It is contained in $[c_i - g, c_i + n + g]$. Since the pair-sum separation hypothesis applied to $2c_i$ and $c_i + c_{i+1}$ implies $c_{i+1} - c_i \geq \Delta \geq n + 2g + 2$, the cross-sum packets are separated from one another by at least one missing integer.

The same-colour X -sum $2P + X + X$ has no nontrivial interval, since $X + X$ lies in one residue class modulo a . Likewise each translate of $Y + Y$ lies in one residue class modulo b and hence has no nontrivial interval. All translates of $Y + Y$ are contained in intervals of the form

$$c_i + c_j - 2g - 2P + [0, 2Sb].$$

The hypothesis that distinct elements of $C + C$ are separated by at least $2Sb + 2$ keeps these $Y + Y$ packets separated from one another by at least one missing integer. Therefore no interval of length n can arise from the same-colour sums or by joining two different packets. The only intervals $[x, x+n] \subseteq 2A$ are the displayed intervals $[c_i, c_i + n]$.

Finally,

$$|A| \leq (R+1) + q(S+1) \leq 3\sqrt{n} + 1 + 2q\sqrt{n},$$

by the estimates already proved in Theorem 3. □

Theorem 5 (Sparse multi-intervals with incompatible colours). *Let $q \geq 1$, let $p_1, \dots, p_q$ be distinct primes at least 5, and put*

$$W = 6 \prod_{i=1}^q p_i.$$

Let $n \geq 1$ satisfy

$$\lfloor \sqrt{n} \rfloor \geq 4W,$$

and let

$$C = \{c_1 < \dots < c_q\} \subseteq \mathbb{Z}$$

with

$$c_{i+1} - c_i \geq 3n \quad (1 \leq i < q).$$

Then there is a finite set $A \subseteq \mathbb{Z}$ such that the intervals $[x, x+n] \subseteq 2A$ are exactly the intervals $[c_i, c_i+n]$, and

$$|A| \leq q(5\sqrt{n} + 1).$$

In particular, for each fixed q this gives a starts-only multi-interval construction with $O_q(\sqrt{n})$ elements and without any pair-sum separation assumption on $C + C$; uniformly, the displayed bound is $O(q\sqrt{n})$.

Proof. Put $N = \lfloor \sqrt{n} \rfloor$. For each i choose, by the Chinese remainder theorem, the unique integer $a_i \in [N - W, N - 1]$ satisfying

$$\begin{aligned} a_i &\equiv 0 \pmod{2}, \\ a_i &\equiv -1 \pmod{3}, \\ a_i &\equiv -1 \pmod{p_i}, \\ a_i &\equiv 0 \pmod{p_j} \quad (j \neq i). \end{aligned}$$

The interval $[N - W, N - 1]$ contains one representative of every residue class modulo W . Since $N \geq 4W$, we have $a_i \geq 3N/4$; and since $N = \lfloor \sqrt{n} \rfloor$, this gives

$$\frac{\sqrt{n}}{2} \leq a_i \leq \sqrt{n} - 1.$$

Put $b_i = a_i + 1$. Then $\gcd(a_i, b_i) = 1$, all a_i are even, all b_i are divisible by 3, and for $i \neq j$ the two integers a_i and b_j are both divisible by p_j .

For each i , repeat the parameter choice from Theorem 3 with a_i in place of a . Thus

$$g_i = a_i(a_i - 1), \quad n_{0,i} = n - (2a_i - 1),$$

choose $v_i \in \{0, \dots, a_i - 1\}$ with $v_i \equiv n_{0,i} \pmod{a_i}$, and put

$$u_i = \frac{n_{0,i} - (a_i + 1)v_i}{a_i}, \quad R_i = a_i + u_i, \quad S_i = a_i - 1 + v_i.$$

Because $a_i + 1 \leq \sqrt{n}$, the same estimates as before give $u_i \geq 0$, $R_i \geq b_i - 1$, $S_i \geq a_i - 1$, and

$$a_i R_i + b_i S_i - 2g_i = n, \quad g_i < n.$$

Let

$$X_i = \{0, a_i, 2a_i, \dots, R_i a_i\}, \quad Y_i = \{0, b_i, 2b_i, \dots, S_i b_i\}.$$

Then Lemma 2 shows that $c_i - g_i + X_i + Y_i$ has the unique interval of length n equal to $[c_i, c_i + n]$.

It remains to keep all unmatched colour sums from interacting. Choose positional labels $\ell_i = M4^{i-1}$ with M so large that packets carrying different nonzero label vectors are separated from one another and from all zero-label packets by at least one missing integer. To make this precise, give every element of X_i

the label vector e_i and every element of Y_i the label vector $-e_i$. Thus the two-term packet $X_i + X_j$ has label $e_i + e_j$, the packet $X_i + Y_j$ has label $e_i - e_j$, and the packet $Y_i + Y_j$ has label $-e_i - e_j$. The only zero-label packets are the matched packets $X_i + Y_i$.

Let T_0 be the maximum absolute value of an element occurring in any unlabelled two-term packet

$$X_i + X_j, \quad c_j - g_j + X_i + Y_j, \quad c_i + c_j - g_i - g_j + Y_i + Y_j,$$

and take $M = 2T_0 + 2$. Lemma 1, with $B = 4$, implies that distinct two-term label vectors of ℓ^1 norm at most 2 differ in their label contribution by at least M , and the unlabelled offsets can change a distance by at most $2T_0$. For the three sign patterns above, a nonzero two-term label vector determines the relevant packet type and its indices, up to the harmless order of the two summands in $X_i + X_j$ or $Y_i + Y_j$.

Define

$$A = \bigcup_{i=1}^q (\ell_i + X_i) \cup \bigcup_{i=1}^q (c_i - g_i - \ell_i + Y_i).$$

The matched sums $(\ell_i + X_i) + (c_i - g_i - \ell_i + Y_i)$ are precisely the zero-label packets considered above. These packets lie in $[c_i - g_i, c_i + n + g_i]$. Since $g_i < n$ and $c_{i+1} - c_i \geq 3n$, distinct zero-label packets are separated by at least one missing integer.

Every unmatched packet has no adjacent elements. Indeed, $X_i + X_j$ lies in one residue class modulo 2, $Y_i + Y_j$ lies in one residue class modulo 3, and, for $i \neq j$, $X_i + Y_j$ lies in one residue class modulo p_j . Because different nonzero label packets are separated from one another and from the zero-label packets, no interval of length n can be formed by an unmatched packet or by joining different packets. Thus the only intervals $[x, x + n] \subseteq 2A$ are the prescribed intervals $[c_i, c_i + n]$.

Finally,

$$|A| \leq \sum_{i=1}^q ((R_i + 1) + (S_i + 1)).$$

Since $a_i \geq \sqrt{n}/2$ and $a_i \leq \sqrt{n}$, the estimates used in Theorem 3 give

$$R_i + 1 \leq 3\sqrt{n} + 1, \quad S_i + 1 \leq 2\sqrt{n}.$$

Summing over i proves the cardinality bound. $\square$

Remark 2. *The pair-sum separation hypothesis in Theorem 4 is exactly what prevents the same-colour sums $Y_i + Y_j$ from interlacing. Thus that theorem has a smaller construction and a milder large- n threshold when the additional additive separation is available. The CRT construction in Theorem 5 removes that additive hypothesis by assigning incompatible colours to different starts, but it asks for larger spacing between the prescribed intervals and has constants depending on q . Neither sparse argument replaces Theorem 2 for complete isolation and arbitrary placement of the starts.*

5 A polynomial diameter refinement

The power labels in Theorem 2 make the first construction transparent, but their size is exponential in the number of target points. The following finite moment encoding gives a polynomial bound for fixed h .

Lemma 3 (Moment separation). *Let $m, L \geq 1$, put*

$$R = \max(1, m - 1), \quad K = LR^{L-1}, \quad Q = LK = L^2R^{L-1},$$

and define

$$b_r = \sum_{j=0}^{L-1} r^j Q^j \quad (0 \leq r < m).$$

If $z = (z_0, \dots, z_{m-1}) \in \mathbb{Z}^m$ is nonzero and $\|z\|_1 \leq L$, then

$$\left| \sum_{r=0}^{m-1} z_r b_r \right| \geq 1.$$

Proof. For $0 \leq j < L$, put

$$\mu_j = \sum_{r=0}^{m-1} z_r r^j.$$

Not all μ_j vanish. Indeed, if the support of z has size s , then $s \leq L$, and the first s equations $\mu_j = 0$ would form an invertible Vandermonde system on the distinct support indices.

Since $|\mu_j| \leq LR^{L-1} = K$, the moment vector has ℓ^1 norm at most $LK = Q$. A nonzero integer coefficient vector with ℓ^1 norm at most Q cannot evaluate to zero in base Q : if J is its largest nonzero index, the leading term has absolute value at least Q^J , while all lower terms together have absolute value at most $(Q-1)Q^{J-1}$ when $J \geq 1$; the case $J = 0$ is immediate. Applying this positional domination to $(\mu_0, \dots, \mu_{L-1})$ proves the claim. $\square$

Theorem 6 (Polynomial diameter realization). *Let $h \geq 2$, $d \geq 2$, and let $S \subseteq \mathbb{Z}$ be a nonempty finite set with $m = |S|$. There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2m$ such that*

$$hA = S \dot{\cup} E,$$

where

$$|x - y| \geq d$$

whenever $x, y \in E$ are distinct or $x \in E$ and $y \in S$.

More precisely, let $\tau \in h\mathbb{Z}$ minimize

$$T = \max_{t \in S} |t - \tau|.$$

Put

$$L = 2h(h-1), \quad R = \max(1, m-1), \quad Q = L^2 R^{L-1},$$

and

$$P_h(m) = \sum_{j=0}^{L-1} (m-1)^j Q^j.$$

Then A can be chosen with

$$\text{diam}(A) \leq h(2hT + d)P_h(m) + T.$$

For fixed h , the factor $P_h(m)$ is bounded by a polynomial in m of degree $L(L-1)$.

Proof. Enumerate $S = \{t_0, \dots, t_{m-1}\}$ and use the parameters in Lemma 3. Set

$$M = 2hT + d, \quad a_r = M \sum_{j=0}^{L-1} r^j Q^j,$$

and first construct the centered set

$$A_0 = \{t_r - \tau + (h-1)a_r : 0 \leq r < m\} \cup \{-a_r : 0 \leq r < m\}.$$

Finally, put

$$A = A_0 + \tau/h.$$

If $\lambda \neq \mu$ and $\|\lambda\|_1 + \|\mu\|_1 \leq L$, then $z = \lambda - \mu$ is nonzero and $\|z\|_1 \leq L$. Lemma 3, after multiplication by M , therefore verifies (1) for the centered target set. Proposition 1 gives $|A_0| = 2m$, the matching-label decomposition, and the required distance bounds. Translating A_0 by τ/h translates hA_0 by τ and does not change its diameter.

The labels a_r are strictly increasing. Every element of A_0 lies between $-a_{m-1}$ and $T + (h-1)a_{m-1}$. Consequently,

$$\text{diam}(A) = \text{diam}(A_0) \leq ha_{m-1} + T = h(2hT + d)P_h(m) + T.$$

For $m \geq 2$, the displayed expression for $P_h(m)$ is a polynomial of degree $L(L-1)$; the case $m = 1$ is immediate. $\square$

Corollary 3 (Translation-invariant interval bound). *Under the hypotheses of Theorem 2, let*

$$m = q(n+1), \quad D = c_q + n - c_1.$$

There is a set A with the same isolated interval pattern and $|A| = 2q(n+1)$ whose diameter is polynomial in m for fixed h , with

$$T \leq \left\lceil \frac{D+h}{2} \right\rceil$$

in the bound of Theorem 6. In particular, the bound is independent of the absolute location of the prescribed intervals.

Proof. Apply Theorem 6 to $S = \bigcup_i [c_i, c_i + n]$. A multiple τ of h can be chosen within $h/2$ of the midpoint of $[c_1, c_q + n]$, giving the stated bound for T . $\square$

6 A Bose–Chowla diameter refinement

The finite moment labels are elementary and formalized, but their polynomial degree is not competitive. A classical construction of Bose and Chowla gives small modular B_t sets, and these can be converted into integer labels with no short relations. We record the conversion explicitly because it is the part used by the present construction.

Let G be an abelian group. A set $B \subseteq G$ is a B_t set if equality of two t -term sums from B , with repetitions allowed, forces equality of the two multisets of summands.

Lemma 4 (Modular B_t labels). *Let $t \geq 1$, let $N \geq 1$, and let $\beta_0, \dots, \beta_{m-1} \in \{0, \dots, N-1\}$ have distinct residue classes forming a B_t set in $\mathbb{Z}/N\mathbb{Z}$. Put*

$$\gamma_r = (2t+1)N + \beta_r.$$

If $z = (z_0, \dots, z_{m-1}) \in \mathbb{Z}^m$ is nonzero and $\|z\|_1 \leq 2t$, then

$$\sum_{r=0}^{m-1} z_r \gamma_r \neq 0.$$

Proof. Write $z = z^+ - z^-$, where $z_r^+ = \max(z_r, 0)$ and $z_r^- = \max(-z_r, 0)$. Let

$$r_+ = \sum_r z_r^+, \quad r_- = \sum_r z_r^-.$$

Then $r_+ + r_- \leq 2t$. Suppose, for a contradiction, that $\sum_r z_r \gamma_r = 0$.

If $r_+ \neq r_-$, then the contribution of the common shift $(2t+1)N$ has absolute value at least $(2t+1)N$. The residue contribution has absolute value at most

$$\sum_r (z_r^+ + z_r^-)(N-1) < 2tN,$$

which is too small to cancel the shift contribution. Hence $r_+ = r_- = r$. In particular, $r \leq t$.

Reducing the equality modulo N gives equality in $\mathbb{Z}/N\mathbb{Z}$ between two r -term sums of the residues β_i . Choose any one residue β_0 and pad both sides by $t - r$ copies of it. The B_t property implies that the two original r -term multisets are equal, so $z = 0$, a contradiction. $\square$

Theorem 7 (Bose–Chowla diameter realization). *Let $h \geq 2$, $d \geq 2$, and let $S \subseteq \mathbb{Z}$ be a nonempty finite set with $m = |S|$. Put*

$$H = h(h - 1).$$

For every prime power $Q \geq m$, there is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2m$ such that

$$hA = S \dot{\cup} E,$$

where distinct elements of E are at distance at least d from one another and every element of E is at distance at least d from S . Moreover, if $\tau \in h\mathbb{Z}$ minimizes

$$T = \max_{s \in S} |s - \tau|,$$

then A can be chosen with

$$\text{diam}(A) \leq h(2hT + d)(2H + 2)(Q^H - 1) + T.$$

Consequently, for $m \geq 2$ one may take a prime $Q \leq 2m - 1$, giving

$$\text{diam}(A) \leq h(2hT + d)(2H + 2)((2m - 1)^H - 1) + T.$$

Proof. By the Bose–Chowla theorem [1], there is a B_H set of cardinality Q in the cyclic group $\mathbb{Z}/(Q^H - 1)\mathbb{Z}$. Choose any m of its elements and represent them by integers $\beta_0, \dots, \beta_{m-1} \in [0, Q^H - 2]$. Lemma 4 with $t = H$ shows that

$$\gamma_r = (2H + 1)(Q^H - 1) + \beta_r$$

has no nonzero integer relation of ℓ^1 norm at most $2H$.

Enumerate the centered target set $S - \tau$ as $t_0, \dots, t_{m-1}$ and put

$$M = 2hT + d, \quad a_r = M\gamma_r.$$

The labels a_r satisfy the separation hypothesis (1). Proposition 1 realizes the centered target set. Translating the resulting basis by τ/h restores S and does not change the diameter.

Finally,

$$\max_r \gamma_r \leq (2H + 1)(Q^H - 1) + (Q^H - 2) < (2H + 2)(Q^H - 1),$$

and every centered generator lies between $-\max_r a_r$ and $T + (h - 1)\max_r a_r$. This gives the displayed diameter bound. The final estimate follows from Bertrand's postulate, choosing a prime $Q \leq 2m - 1$ for $m \geq 2$. $\square$

Theorem 8 (Quadratic diameter for double sumsets). *Let $d \geq 2$, and let $S \subseteq \mathbb{Z}$ be a nonempty finite set with $m = |S|$. There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2m$ such that*

$$2A = S \dot{\cup} E,$$

where distinct elements of E are at distance at least d from one another and every element of E is at distance at least d from S .

More precisely, let $\tau \in 2\mathbb{Z}$ minimize

$$T = \max_{t \in S} |t - \tau|,$$

and let $p \geq m$ be an odd prime. For $0 \leq r < m$, let ρ_r be the least nonnegative residue of r^2 modulo p , and put

$$b_r = 2p^2 + 2pr + \rho_r.$$

Then A can be chosen with

$$\text{diam}(A) \leq 8(4T + d)p^2 + T.$$

For $m \geq 2$, Bertrand's postulate allows $p < 2m$, giving a quadratic bound in m . Since p is an integer, the resulting explicit estimate is

$$\text{diam}(A) \leq 8(4T + d)(2m - 1)^2 + T.$$

Proof. First observe that $\{b_0, \dots, b_{m-1}\}$ is a Sidon set, with repetitions allowed in pair sums. Indeed, suppose

$$b_i + b_j = b_k + b_\ell.$$

Reducing modulo $2p$ gives $\rho_i + \rho_j = \rho_k + \rho_\ell$, because both sides lie between 0 and $2p - 2$. It follows that $i + j = k + \ell$. Reducing the residue identity modulo p also gives

$$i^2 + j^2 \equiv k^2 + \ell^2 \pmod{p}.$$

Since p is odd, we obtain $ij \equiv k\ell \pmod{p}$. Thus the two pairs are the roots of the same quadratic polynomial over $\mathbb{F}_p$. The unordered pairs $\{i, j\}$ and $\{k, \ell\}$ agree modulo p , hence agree as integer pairs because all indices lie in $[0, p)$.

Moreover,

$$2p^2 \leq b_r < 4p^2.$$

Consequently, an equality between two sums of the b_r involving at most four terms in total must have the same number of terms on both sides: one term is strictly smaller than any sum of two terms. Equal one-term sums are trivial, and equal two-term sums are trivial by the Sidon property. Therefore

$$\sum_r z_r b_r \neq 0$$

for every nonzero integer vector z with $\|z\|_1 \leq 4$.

Set $M = 4T + d$, enumerate $S = \{t_0, \dots, t_{m-1}\}$, and apply Proposition 1 to the centered targets $t_r - \tau$ and the labels $a_r = Mb_r$. Finally translate the resulting basis by $\tau/2$. Since $b_r < 4p^2$, its diameter is at most

$$2M \max_r b_r + T \leq 8(4T + d)p^2 + T.$$

The same construction has quadratic diameter from below. Indeed, translation does not change diameter, the negative generators contain $-Mb_0$ and $-Mb_{m-1}$, and

$$b_{m-1} - b_0 = 2p(m - 1) + \rho_{m-1} \geq 2m(m - 1).$$

Consequently,

$$\text{diam}(A) \geq 2Mm(m - 1).$$

□

Proposition 3 (A Sidon-label diameter barrier). *Let $B = \{b_1, \dots, b_m\} \subseteq \mathbb{Z}$ be a Sidon set, where repetitions are allowed in pair sums. Then*

$$\text{diam}(B) \geq \binom{m}{2}.$$

Consequently, any tagged construction that contains the generators $-Mb_i$ has diameter at least $M\binom{m}{2}$.

Proof. Order the labels so that $b_1 < \dots < b_m$. The $\binom{m}{2}$ positive differences $b_j - b_i$, with $i < j$, are distinct. Indeed, an equality $b_j - b_i = b_\ell - b_k$ gives $b_j + b_k = b_\ell + b_i$, and the Sidon property forces the two ordered index pairs to agree. Every positive difference is at most $\text{diam}(B)$, proving the first claim. The second follows because the generators $-Mb_i$ already have diameter $M \text{diam}(B)$. $\square$

Remark 3 (Elementary lower bounds). *Let $A \subseteq \mathbb{Z}$ be finite and nonempty, put $k = |A|$, and suppose that $S \subseteq hA$ has m elements. Then*

$$m \leq \binom{k+h-1}{h}, \quad \text{diam}(A) \geq \frac{\text{diam}(S)}{h}.$$

Indeed, the first inequality counts unordered h -term multisets from A , and the second follows from

$$\text{diam}(hA) = h \text{diam}(A).$$

Thus Theorem 6 supplies an elementary polynomial diameter bound for fixed h , while Theorem 7 lowers the exponent to $H = h(h-1)$ by using classical modular B_H sets. Theorem 8 gives a quadratic bound when $h = 2$ with better constants and a direct integer formula. Proposition 3 shows that the quadratic order is unavoidable within the Sidon-label method. Optimizing the cardinality for the arbitrary multi-interval isolated problem, or proving stronger lower bounds outside these label frameworks, remains open.

7 Relation with the single-interval construction

Nathanson’s version 3 contains a direct construction for one prescribed interval in a double sumset. The special case $q = 1$ and $h = 2$ of Corollary 1 gives another route to the same type of single-interval conclusion, but this is not the main point of the present paper. The label construction is designed to make independent cancellations for many target points coexist without producing additional adjacent exceptional sums. This is what allows the same argument to treat the multi-interval questions and the higher-fold case uniformly.

The quantitative refinements also belong to the finite-target viewpoint. Moment labels make the construction polynomial in the number of target points for fixed h , after centering the target set. For $h = 2$, shifted Sidon labels give a quadratic diameter bound, and Proposition 3 shows that this order is forced by the Sidon-label mechanism itself.

The sparse construction in Theorem 3 points in a different direction: it optimizes cardinality rather than isolation. It shows that the $2n+2$ size in the isolated single-interval specialization is not a lower bound for the starts-only single-interval problem. Proposition 2 shows that the resulting $O(\sqrt{n})$ cardinality is optimal up to the absolute constant. Theorem 4 extends the same packet to several starts when the pair-sum set $C + C$ is separated. The incompatible-colour construction in Theorem 5 removes the pair-sum condition for fixed q and large n , at the cost of a larger spacing hypothesis. Theorem 2 remains the arbitrary-placement and complete-isolation result.

8 Reproducibility

The accompanying Python package constructs A , computes hA , checks all maximal consecutive runs, records the minimum exceptional spacing, and emits JSON certificates. It also includes exact small-parameter computations cross-checked against [OEIS A001208](#) and [OEIS A001209](#), together with bounded scans for sets whose double sumset has exactly one nontrivial interval. These bounded scans are used only as discovery evidence for cardinality questions. The arithmetic separation, bounded-offset, matched-count, tagged-representation, exceptional-count uniqueness, generic set-level isolation, interval-enumeration, maximal-run, and exact-cardinality arguments have also been checked in Lean 4 with Mathlib. The resulting Lean theorem automatically selects the target radius and substitutes the displayed

values of M and B . The finite moment-label Vandermonde argument, translation rule, closed diameter estimate, exact cardinality, midpoint-centering lemma, and interval-family theorem have also been formalized. For the shifted-Sidon refinement, Lean checks the size bounds and the reduction from order-four short-relation freedom to set-level isolation, including the explicit pair-sum argument over $\mathbb{F}_p$. It also checks translation invariance, quadratic diameter upper and lower bounds, tightened Bertrand-prime selection, exact cardinality after translation, and the final separated-interval wrapper. The midpoint translation, radius, and scale are selected automatically from an enclosing interval. The sparse Frobenius, pair-sum-separated, and CRT-incompatible constructions are implemented in the Python package and are included as ordinary paper proofs, not as Lean theorems.

References

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